Longitudinal Data Analysis

Homework 3

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1 Introduction

1.1 Objectives

This study aims to investigate how hemoglobin (Hb) status changes over time among patients with renal deficiency undergoing hemodialysis. The study will assess the influence of iron deficiency status and demographic factors, including age and sex. Additionally, we will compare multiple analysis methods as well as try to explore if the patients' measurements were missing at any time points and how those could influence our results.

2 Data Management and Exploration

2.1 Data Management

The dataset includes monthly measurements collected over a six-month follow-up period from patients undergoing hemodialysis. These measurements include hemoglobin (Hb, in g/dL), categorized as a binary outcome as welll as a continuous outcome. The binary outcome is defines as severe (Hb ≤ 10) vs. not severe anemia (Hb > 10). Additional variables included the erythropoietin (EPO) dose (IU/kg/week, IU: International Unit, discrete), and iron deficiency status (categorical, defined as present if serum ferritin levels are below 100 μg/L and/or transferrin saturation is below 20%). Additionally, demographic data, including sex (categorical) and age (discrete), were collected.

As in previous analysis, the lagged variables for EPO dose and Hb were used to acocunt for delayed effects of EPO on Hemoglobin outcomes over time, reflecting the clinical adjustments based on prior values rather than immediate changes. Iron deficiency was analyzed as observed during the same month, given its rapid treatment response and immediate relevance in dosing decisions. Missing data, which was previously not considered for analysis, was again observed and required further exploration which will be described in the following section.

3 Data Analysis - Missingness

An overview of the missing data present accross all variables is shown in 1. The response variable had 395 missing values. Dose and Sex had 463 and 342 respectively. It should be noted that the missingness in Sex as a time-invariant covariate was not expected and should be explored further. There were also 345 occassions where subjects did not have a recorded month, due to them being missing from the study or a missingness problem when recording them.
The missingness in the lagged variables reflects dependencies on prior months' data, where missingness in earlier months propagates into lagged variables. It also presents structural missingness due to them being an artifact of the original data and missing the observation at Month 1 but this pattern is not stochastic and therefore should not affect our modeling as long as the variables are accounted for.

Table 1: Missingness in all covariates

	Hb (Binary)	**Dose**	**Sex**	**Age**	**Iron**	**Lag_Hb**	**Lag_Dose**	**Month**
Missing	395	463	342	0	0	3894	3856	345

To further analyze the missingness on our dataset we adopted the standard taxonomy described on Little and Rubin (1987). For each subject i, the sequence of measurements Y_{ij} at each time point j was used to construct the vectors $R_i = (R_{i1}, \ldots, R_{in_i})$ which capture the missingness in each subject i at time points $j = 1, \ldots, n_i$:

$$R_{ij} = \begin{cases} 1 & \text{if } Y_{ij} \text{ is observed,} \\ 0 & \text{otherwise.} \end{cases}$$

This framework allows us to distinguish between the observed (Y_i^o) and missing components (Y_i^m) to be able to determine which pattern is followed by each patient. As seen in Table 2, only a group of 28 patients presented measurements at all time points in the study and are therefore called Completers, six groups patients dropped out of the study after a certain time point, therefore following a Monotone Missingness pattern, while the rest are present at intermittent time points, which is denoted as Non Monotone Missingness. While monotone missigness is easier to handle as it progresses naturally over time, Non-monotone missingness is more complex and requires additional attention and modeling considerations.

Table 2: Missingness Patterns by Measurement Occasion

Measurement Occasion	Pattern	Count
Completers		
Month 1–6	O O O O O O	28
Dropouts (Monotone Missingness)		
Month 1–6	O O O O O M	121
Month 1–6	O O O O M M	734
Month 1–6	O O O M M M	867
Month 1–6	O O M M M M	626
Month 1–6	O M M M M M	435
Month 1–6	M M M M M M	349
Non-Monotone Missingness		
Month 1–6	M M M M O M	1
Month 1–6	M M M O M M	10
Month 1–6	M M M O O M	1
Month 1–6	M M M O O O	2
Month 1–6	M M O M M M	32
Month 1–6	M M O O M M	26
Month 1–6	M M O O O M	8
Month 1–6	M O M M M M	63
Month 1–6	M O M O M M	9
Month 1–6	M O M O O M	1
Month 1–6	M O O M M M	57
Month 1–6	M O O M O M	1
Month 1–6	M O O O M M	95
Month 1–6	M O O O O M	7
Month 1–6	O M M M O O	1
Month 1–6	O M M O M M	12
Month 1–6	O M M O O M	6
Month 1–6	O M O M M M	113
Month 1–6	O M O O M M	63
Month 1–6	O M O O O M	7
Month 1–6	O M O O O O	2
Month 1–6	O O M O M M	120
Month 1–6	O O M O O M	11
Month 1–6	O O M O O O	4
Month 1–6	O O O M O M	11

3.1 Missingess Mechanism - MAR

As the missing data mechanism governs the choice of analysis, we further explored the dropout vectors R_i to determine if a pattern of missingness in the unobserved components (Y_i^m) could be found. Ideally no patterns should be observed so that the Missing Completely at Random (MCAR) assumption would hold and the data could be analyzed

without additional steps or biases. However, in practice, data is often Missing At Random (MAR) or Missing Not At Random (MNAR). That is, conditional on the observed data or covariates or conditional on the unobserved values, respectively.
The probability of missingness in the binary response variable (HBBIN) was modeled using logistic regression on the missingness indicator vector R_i as a function of the observed covariates. The results indicated that the missingness in HBBIN was significantly associated with Sex ($p = 0.044$), a variable that itself had unobserved values, and the second month of observation ($p = 0.040$). The same process was repeated for the covariate Sex, which is time-invariant and therefore was found to only depend on Age ($p = 0.0039$), the other baseline covariate, which was fully observed. For probability of missingness in dose we found an association with the second month ($p = 0.032$) as well as the response variable HBBIN ($p = 0.039$).his result aligns with clinical expectations, as the dose administered to patients depends on the measured hemoglobin levels, and missingness could arise from differences in treatment.
Overall, we couldn't justify the MAR assumption for this dataset as is to directly use some of the methods described in the following sections, as missingness on the response does not depend on fully observed variables directly, and there is also missingness in the covariates. So, to address the potential bias introduced by missingness and to enable consistent comparisons between methods using the same starting dataset, we performed Multiple Imputation on the missing data. This ensured that all methods were valid under the MAR assumption.

4 Material and Methods

4.1 Linear Mixed Model (LMM)

The linear mixed model (LMM) is a powerful statistical approach for analyzing longitudinal data. The model incorporates both fixed and random effects to account for the correlation within subjects' measurements over time. The fixed effects represent the population-level parameters, while the random effects account for individual variations between subjects.

The model is expressed as follows:

$$Y_i = X_i\beta + Z_i b_i + \varepsilon_i \tag{1}$$

In this equation, Y_i represents the vector of repeated hemoglobin (Hb) measurements for the i-th subject. X_i is the matrix of covariates, and β is the vector of fixed-effect parameters. Z_i is the design matrix for the random effects, and b_i denotes the random intercepts for each subject. ε_i represents the residual error term. It is assumed that the random intercepts b_i follow a normal distribution, $b_i \sim N(0, D)$, where D is the variance-covariance matrix for the random effects. The residuals ε_i are also assumed to follow a normal distribution, $\varepsilon_i \sim N(0, \Sigma_i)$, where Σ_i represents the correlation structure within each subject over time.

For the current analysis, we adopted the most parsimonious mean structure, which was determined in our earlier report through a backward selection method, supported by a thorough exploration of the data to guide the choice of the appropriate variance-covariance structure.

$$\begin{aligned}\text{Hb}_{it} = \beta_0 &+ \beta_1\text{Age}_i + \beta_2\text{Sex}_i + \beta_3\text{Time}_{it} + \beta_4\text{Time}_{it}^2 \\ &+ \beta_5\text{Iron}_{it} + \beta_6\text{LagDose}_{it} + \beta_7\text{LagHb}_{it} \\ &+ \beta_8(\text{Time}_{it} \times \text{Age}_i) + \beta_9(\text{Time}_{it} \times \text{Iron}_{it}) \\ &+ \beta_{10}(\text{Time}_{it}^2 \times \text{Iron}_{it}) + \beta_{11}(\text{Time}_{it} \times \text{LagDose}_{it}) \\ &+ b_{0i} + \epsilon_{it}\end{aligned}$$

Where:

- Y_{it} is the repeated measurement (i.e., Hb levels) for subject i at time t,
- $\beta_0, \beta_1, \ldots, \beta_{11}$ are the fixed effects coefficients,
- b_{0i} is the random intercept for subject i,
- ϵ_{it} is the residual error term,
- LagDose_{it} and LagHb_{it} represent the lagged values of the dose and hemoglobin level, respectively.

The random intercept b_{0i} allows for the modeling of individual variability in hemoglobin levels that is not explained by fixed effects. This helps to account for the fact that individuals may have baseline differences in hemoglobin levels, which is important for accurate predictions.

Given the highly unbalanced nature of the dataset, where many subjects have missing data points across different time points (and only 27 subjects completed the full follow-up period), we could not rely solely on direct maximum likelihood (ML) or restricted maximum likelihood (REML) estimation methods. While REML is robust in unbalanced datasets by estimating the random effects more reliably, it assumes that missing data are either missing at random (MAR) or missing completely at random (MCAR). However, in the case of missing covariates, REML or direct likelihood methods would not be sufficient on their own, as they do not fully account for the uncertainty introduced by missing values in the covariates.

To address the issue of missing data, we opted for multiple imputation, a robust method that generates multiple plausible datasets by filling in the missing covariate values. This approach not only handles the missing data but also incorporates the uncertainty associated with the missing values, which helps mitigate bias in the parameter estimates.

An alternative approach to imputing missing data is through conditional densities. Fully Conditional Specification (FCS) imputes missing values for each variable individually, using a specified imputation model for each incomplete variable and performing imputations iteratively (Van Buuren, 2018). In our analysis, we implemented FCS in SAS, with 10 imputations (m=10). Continuous variables, such as hb and dose, were imputed using standard regression models, while logistic regression models were applied to categorical variables, including iron and sex. We incorporated variables identified in the dropout analysis, such as doseprev and age, into each imputation model (Molenberghs and Kenward, 2007). For the ordinal outcome, we first imputed the continuous data before transforming it into its categorized version.

Linear mixed model analysis was performed using PROC MIXED, with imputation handled by PROC MI and subsequent analysis conducted with PROC MIANALYZE in SAS.

4.2 Marginal Model - Doubly Robust Estimating Equations (DRGEE)

For this analysis, the Doubly Robust Generalized Estimating Equation (DR-GEE) approach was chosen to fit a marginal model under the Missing at Random (MAR) assumption. Prior to applying DR-GEE, Multiple Imputation (MI) was performed to handle missing values. MI generated multiple complete datasets, preserving variability introduced by missing data while utilizing all available information under the MAR assumption. This approach ensured that the analysis accounted for uncertainty in the imputed values. DR-GEE was subsequently applied because of its robustness to model misspecification. It provides valid and reliable estimates as long as either the outcome model or the propensity score model is correctly specified, making it a powerful tool in the presence of missing data and potential confounding. To adjust for differences between observed and missing data, inverse probability weighting (IPW) was employed, using propensity scores (pscore) estimated via logistic regression based on covariates associated with the binary hemoglobin outcome ($HBBIN$) and predictors of missingness. This two-step approach effectively combines the strengths of MI and IPW, ensuring a more representative analysis of the population.

The weights were calculated as:

$$\text{Weight} = \begin{cases} \frac{1}{\text{pscore}}, & \text{if } HBBIN = 1, \\ \frac{1}{1-\text{pscore}}, & \text{if } HBBIN = 0, \end{cases}$$

where pscore represents the estimated propensity score from the logistic regression model:

$$\begin{aligned} \log\left(\frac{\text{pscore}}{1-\text{pscore}}\right) = \beta_0 &+ \beta_1 \cdot \text{SEX} + \beta_2 \cdot \text{AGE} \\ &+ \beta_3 \cdot \text{IRON} + \beta_4 \cdot \text{HBLAG} + \beta_5 \cdot \text{DOSELAG} \\ &+ \beta_6 \cdot \text{TIME} + \beta_7 \cdot \text{TIME}^2 \\ &+ \beta_8 \cdot (\text{HBLAG} \cdot \text{TIME}) + \beta_9 \cdot (\text{DOSELAG} \cdot \text{TIME}) \\ &+ \beta_{10} \cdot (\text{TIME}^2 \cdot \text{HBLAG}). \end{aligned}$$

Where:

- β_0: Baseline log odds of $HBBIN = 1$ when all covariates are zero.
- β_1: Effect of being male ($SEX = 1$) on the log odds of $HBBIN = 1$.
- β_2: Effect of a one-year increase in age on the log odds of $HBBIN = 1$.
- β_3: Effect of a one-unit increase in iron level on the log odds of $HBBIN = 1$.
- β_4: Effect of the lagged hemoglobin ($HBLAG$) on the log odds of $HBBIN = 1$.
- β_5: Effect of the lagged dose ($DOSELAG$) on the log odds of $HBBIN = 1$.
- β_6: Linear effect of time on the log odds of $HBBIN = 1$.
- β_7: Quadratic effect of time (TIME^2) on the log odds of $HBBIN = 1$.

- β_8: Interaction effect of lagged hemoglobin ($HBLAG$) and time on the log odds of $HBBIN = 1$.
- β_9: Interaction effect of lagged dose ($DOSELAG$) and time on the log odds of $HBBIN = 1$.
- β_{10}: Interaction effect of quadratic time (TIME^2) and lagged hemoglobin ($HBLAG$) on the log odds of $HBBIN = 1$.

To stabilize variance and reduce the influence of extreme weights, the weights were normalized as follows:

$$\text{Normalized Weight} = \frac{\text{Weight}}{\text{Mean Weight}},$$

ensuring comparability and stability in the analysis.

The final DR-GEE model was specified as:

$$\text{logit}(\Pr(HBBIN = 1)) = \alpha_0 + \alpha_1 \cdot \text{TIME} + \alpha_2 \cdot \text{AGE} + \alpha_3 \cdot \text{SEX} + \alpha_4 \cdot \text{IRON} + \alpha_5 \cdot \text{DOSELAG} + \alpha_6 \cdot \text{HBLAG}$$

$$+ \alpha_7 \cdot \text{TIME}^2 + \alpha_8 \cdot (\text{TIME} \cdot \text{DOSELAG}) + \alpha_9 \cdot (\text{TIME} \cdot \text{HBLAG}) + \alpha_{10} \cdot (\text{TIME}^2 \cdot \text{HBLAG}),$$

where:

- α_0: Intercept, representing the log odds of $HBBIN = 1$ when all covariates are zero.
- α_1: Linear effect of time on the log odds of $HBBIN = 1$.
- α_2: Effect of a one-year increase in age on the log odds of $HBBIN = 1$.
- α_3: Effect of being male ($SEX = 1$) on the log odds of $HBBIN = 1$.
- α_4: Effect of a one-unit increase in iron levels ($IRON$) on the log odds of $HBBIN = 1$.
- α_5: Effect of the lagged dose ($DOSELAG$) on the log odds of $HBBIN = 1$.
- α_6: Effect of the lagged hemoglobin ($HBLAG$) on the log odds of $HBBIN = 1$.
- α_7: Quadratic effect of time (TIME^2) on the log odds of $HBBIN = 1$.
- α_8: Interaction effect of time and lagged dose ($TIME \cdot DOSELAG$).
- α_9: Interaction effect of time and lagged hemoglobin ($TIME \cdot HBLAG$).
- α_{10}: Non-linear interaction effect of quadratic time and lagged hemoglobin ($TIME^2 \cdot HBLAG$).

The DR-GEE model incorporates normalized weights (norm_weight) to adjust for confounding and ensure doubly robust estimation. The weights are applied during the estimation process to solve the weighted estimating equations:

$$\sum_i \text{norm_weight}_i \cdot \mathbf{X}_i \cdot [HBBIN_i - \mu_i(\mathbf{X}_i, \boldsymbol{\alpha})] = 0,$$

where:

- $\mathbf{X}_i$: Covariates for the i-th observation.
- μ_i: Expected value of $HBBIN_i$ under the model.
- $\boldsymbol{\alpha}$: Coefficient vector.

4.3 Generalized Linear Mixed Models (GLMM)

Generalized Linear Mixed Models (GLMMs) extend Generalized Linear Models (GLMs) by incorporating random effects to model subject-specific variability. This model is suited for this particular case due to the longitudinal nature of the data and the binary response, modeled through a logit link function. The logit link function transforms the probability of the binary outcome (Y = 1) into a linear combination of predictors, making it compatible with the GLMM framework and making sure the model produces interpretable results on the odds scale.
The GLMM model equation is specified as:

$$\text{logit}(\Pr(Y_{ij} = 1)) = \alpha + \sum_{k=1}^{p} X_{ijk}\beta_k + b_{i0} + b_{i1} \cdot \text{Time}_j.$$

where:

- $\text{logit}(\Pr(Y_{ij} = 1))$: The log-odds of the binary outcome Y_{ij} for subject i at time j.
- α: Fixed intercept representing the baseline log-odds of $Y = 1$ when all covariates are zero.
- $\sum_{k=1}^{p} X_{ijk}\beta_k$: Fixed effects capturing the population-level relationship between the predictors (X_{ijk}) and the log-odds of the binary outcome, with coefficients β_k.
- b_{i0}: Random intercept capturing subject-specific deviations from the population-level intercept α.
- b_{i1}: Random slope capturing subject-specific deviations in the effect of time (Time_j) on the log-odds.

The random effects structure is specified as:

$$\mathbf{b}_i = \begin{bmatrix} b_{i0} \\ b_{i1} \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_u^2 & \text{Cov}(u,v) \\ \text{Cov}(u,v) & \sigma_v^2 \end{bmatrix} \right),$$

where:

- b_{i0}: Random intercept for subject i, representing deviations from the fixed intercept α.
- b_{i1}: Random slope for subject i, capturing deviations in the effect of time (Time_j) on the log-odds.
- σ_u^2: Variance of random intercepts (b_{i0}).
- σ_v^2: Variance of random slopes (b_{i1}).
- $\text{Cov}(u, v)$: Covariance between random intercepts and slopes.

Missing data were handled through Multiple Imputation (MI) using a Markov Chain Monte Carlo (MCMC) method, generating multiple plausible complete datasets consistent with the observed data under the Missing At Random (MAR) assumption. While MI is a powerful tool for handling missing data, direct likelihood-based methods, such as those used in GLMMs, could also provide valid estimates under the MAR assumption without requiring imputation. These methods maximize the likelihood over the observed data directly, appropriately accounting for missingness in the response and are naturally compatible with the GLMM framework. However, due to the missingess in time-varying covariates and to ensure comparability with other methods that do not allow for direct likelihood, we fitted the GLMM to the same imputed datasets. The GLMM was fitted to each imputed dataset using the SAS procedure `PROC GLIMMIX`, and the results were pooled using `PROC MIANALYZE` to obtain valid parameter estimates under Rubin's rules.

5 Results and Discussion

5.1 Linear Mixed Model (LMM)

Significantly different results were obtained through multiple imputation compared to when relying on direct likelihood. Briefly, we previously found that iron status had a strong non linear HB response over time (0.2971, $p = 0.0064$), and significant linear relationships for age (-0.00186, $p = 0.0263$) and significant relationship for LagDose over time (0.000335, $p = 0.0174$). Conversely, the analysis with multiple imputation did not provide significant results for categorical variables including sex, iron and their interactions, with estimates 0, meaning that there was no significant variability. Therefore, these variables were removed from the model. In this model, age and lagDose did not have a significant effect on Hb over time, yet high levels of hemoglobin at baseline evidenced a significant positive effect on Hb. Table 3 summarizes the results of the final linear mixed model.

Table 3: Parameter Estimates for the Linear Mixed Model

Parameter	**Estimate**	**95% Confidence Limits**	**P value**
Intercept	3.1968	1.6177 - 4.773045	0.0012
Time	-0.1819	-0.6256 - 0.261780	0.3828
Age	-0.0017	-0.0075 - 0.004172	0.5572
LagDose	0.0014	-0.0012 - 0.0050	0.5572
LagHb	0.7285	0.6139 - 0.8431	¡0.0001
Time*Age	0.0005	-0.0012 - 0.0022	0.5293
Time*LagDose	-0.0002	-0.0014 - 0.0009	0.7296

In contrast with the direct likelihood model, in which the Gaussian serial correlation explained most of the variability of Hb, the covariance estimates obtained after multiple imputation show that the variability of Hb over time is mostly explained by the random intercept, followed by measurement error.

Table 4: Covariance Parameter Estimates for the Linear Mixed Model

Cov Param	Estimate
UN(1,1)	7.74E-19
Variance (Random Intercept)	0.6739
SP(GAU)	2.42E-17
Residual	0.5016

5.2 Marginal Model - Doubly Robust Estimating Equations (DRGEE)

Propensity Score Model

Table 5: Propensity Score Logistic Regression Results

Variable	Estimate	Standard Error	Wald Chi-Square	p-value
Intercept	5.6283	0.6578	73.1997	<0.0001
Sex (1 vs. 2)	0.00745	0.0134	0.3089	0.5784
Age	-0.00135	0.000445	9.2168	0.0024
Iron (0 vs. 1)	-0.0471	0.0151	9.7767	0.0018
Lagged Hemoglobin (laghb)	-0.5820	0.0635	83.9539	<0.0001
Lagged Dose (lagdose)	-0.00179	0.000237	56.9284	<0.0001
Time	5.0950	0.3534	207.9015	<0.0001
Time^2 (time*time)	-0.7789	0.0428	330.4538	<0.0001
LagHB $\times$ Time	-0.5135	0.0342	225.3878	<0.0001
LagDose $\times$ Time	0.000347	0.000054	41.4734	<0.0001
LagHB $\times$ Time^2	0.0778	0.00415	352.2343	<0.0001

The propensity score model estimates the likelihood of hemoglobin levels being ≤ 10 based on selected covariates and demonstrated excellent predictive performance with a c-statistic of 0.834, indicating for 83.4 percent of all possible pairs of observations (where one observation has hemoglobin levels less than or equal to 10 and the other has levels upper than 10), the model correctly predicts a higher propensity score for the observation with hemoglobin levels ≤ 10. Sex was not statistically significant ($p = 0.5784$), indicating no notable difference between sexes in predicting hemoglobin levels. Lagged hemoglobin levels were strongly associated with lower odds of hemoglobin ≤ 10 ($OR = 0.559$), meaning that higher prior hemoglobin levels reduced the likelihood of low hemoglobin. Similarly, higher lagged dose was associated with slightly lower odds ($OR = 0.998$), although the effect was small. Time had a significant positive association ($OR = 163.68$), suggesting that the odds of hemoglobin ≤ 10 increased over time, but the quadratic term ($OR = 0.459$) indicated that this increase slowed over time. Interaction terms, particularly between lagged hemoglobin and time ($OR = 0.598$), underscored the importance of temporal trends in predicting hemoglobin levels. These results will be used

to adjust for confounding in subsequent analyses.

Final Model DR GEE Results

Table 6: Final DR-GEE Model Results

Variable	**Estimate**	**Robust SE**	**p-value**
Intercept	0.4356	0.9330	0.6406
Time	-0.3033	0.4869	0.5333
Age	0.0010	0.0013	0.4400
Sex (2 vs. 1)	0.0631	0.0565	0.2639
Iron (1 vs. 0)	0.0146	0.0125	0.2423
Lagged Dose (lagdose)	-0.0001	0.0002	0.3875
Lagged Hemoglobin (laghb)	0.0367	0.0865	0.6713
Time^2 (time*time)	0.0328	0.0518	0.5264
Time $\times$ LagDose	-0.0000	0.0000	0.9970
Time $\times$ LagHB	0.0300	0.0457	0.5122
$\text{Time}^2 \times$ LagHB	-0.0031	0.0048	0.5160

After applying inverse probability weights derived from the propensity score model, the final DR-GEE analysis showed that none of the variables retained statistical significance. This outcome highlights the role of weighting in adjusting for differences in covariate distributions between groups. In the unweighted propensity score model, variables such as lagged hemoglobin, lagged dose, and time were significant predictors of hemoglobin levels. However, after weighting, the adjusted population no longer exhibited strong residual effects from these covariates, suggesting that the observed relationships were primarily due to imbalances in the original data. The DR-GEE framework also accounts for within-subject correlations and temporal variability, which can increase standard errors and further influence statistical significance. These findings indicate that the propensity score weighting effectively balanced the dataset, allowing for a more accurate and unbiased assessment of the relationships between covariates and hemoglobin levels. This ensures the robustness of the analysis in reflecting true population-level effects.

Comparison between DR GEE vs GEE

Table 7: GEE Analysis with Continuous Hemoglobin

Effect	Estimate	SE	Robust SE	p-value
Intercept	14.1314	1.1348	1.2555	<0.0001
Month	-3.3596	0.7494	0.8412	<0.0001
Age	0.0012	0.0014	0.0014	0.3829
Sex (Male)	0.1135	0.0411	0.0411	0.0057
Iron (adequate)	0.1014	0.0304	0.0310	0.0011
Lagged Hb	-0.3007	0.1014	0.1112	0.0069
Lagged Dose	0.0008	0.0005	0.0005	0.1214
Month × Month	0.5189	0.1182	0.1341	0.0001
Month × Lagged Dose	0.0002	0.0017	0.0750	0.2516
Month × Lagged Hb	0.2953	0.0671	0.0750	<0.0001
Month × Month × Lagged Hb	-0.0456	0.0105	0.0120	0.0001

The final DR-GEE model, which used a binary outcome (hemoglobin levels ≤ 10 vs. > 10) and accounted for missing data under the MAR assumption, showed no significant effects for any predictors after applying inverse probability weights. In contrast, the GEE model, which analyzed continuous hemoglobin levels without adjusting for missingness, identified several significant predictors, including sex, iron status, lagged hemoglobin, and their interactions with time. These differences are due to the DR-GEE model adjusting the analysis to reflect a weighted population where covariates are balanced, which reduces the impact of initial imbalances in the data. However, this adjustment increases variability in parameter estimates, leading to wider confidence intervals and reduced statistical significance. Furthermore, the binary nature of the DR-GEE outcome simplifies relationships between predictors and hemoglobin levels by focusing on odds ratios, while the continuous GEE model captures absolute changes, making it more sensitive to subtle effects of covariates. Therefore, with DR-GEE we are ensuring robustness in the presence of missing data and GEE emphasizing direct effects on mean hemoglobin levels.

5.3 Generalized Linear Mixed Models (GLMM)

Initially, the most saturated mean structure was fitted, including main effects for all variables, interactions with month, as well as interactions that were found significant in previous explorations of the dataset. Additionally, we considered the interaction between age and sex, which was shown to jointly affect missing data and was also significant in a previous analysis.

5.3.1 Test for Random Slopes

To determine the random effects structure, an exploratory analysis was conducted using the original dataset without multiple imputation. We tested whether the addition of a

random slope would improve the model fit as compared to a model with only a random intercept. The likelihood ratio test statistic was calculated as $LRT = 3272.102807.94 = 464.16$. Under the null hypothesis that the variance of the random slope is 0, the LRT statistic follows a 50:50 mixture of χ_1^2 and χ_2^2 distributions due to the boundary condition of variance parameters. The p-value was computed as $p = 0.5 \times P(\chi_1^2 > 464.16) + 0.5 \times P(\chi_2^2 > 464.16)$. Therefore, given the large test statistic and small p-value, the addition of a random slope for time was deemed to significantly improve the model fit.

The final model after mean structure reduction and with the same random effects structure was then applied to the multiple imputed datasets to ensure consistency across analyses while leveraging the advantages of multiple imputation for handling missing data. Since direct likelihood approaches were used, the results should be unbiased even without multiple imputation, since all variables that had missingness in them as well as their dependencies were included in the models.

Table 8: Test for Random Slope

Model	**-2log lik**	**Chisq**	**p-value**
Random intercept	3272.10	-	-
Random intercept + slope	2807.94	464.16	<.0001

5.3.2 Final Model

The final model is defined as follows:

$$\begin{aligned}\log\left(\frac{P(\text{HBBIN}_{ij}=1)}{1-P(\text{HBBIN}_{ij}=1)}\right) = {} & \beta_0 + \beta_1 \cdot \text{Age}_i + \beta_2 \cdot \text{Sex}_i + \beta_3 \cdot \text{Iron}_{ij} + \beta_4 \cdot \text{LagDose}_{ij} + \beta_5 \cdot \text{LagHB}_{ij} \\ & + \beta_6 \cdot \text{Month}_{ij} + \beta_7 \cdot (\text{LagHB}_{ij} \cdot \text{Month}_{ij}) + \beta_8 \cdot (\text{Age}_i \cdot \text{Month}_{ij}) \\ & + \beta_9 \cdot (\text{Sex}_i \cdot \text{Month}_{ij}) + \beta_{10} \cdot (\text{LagDose}_{ij} \cdot \text{Iron}_{ij}) \\ & + \beta_{11} \cdot (\text{Age}_i \cdot \text{Sex}_i) + b_{i0} + b_{i1} \cdot \text{Month}_{ij}.\end{aligned} \tag{2}$$

For $i = 1, \ldots, n$ subjects, measured across $j = 1, \ldots, 6$ time points for each subject i:

- β_0: The intercept term, representing the baseline log-odds of having hemoglobin $\leq 10\,\text{g/dL}$ when all predictors are at their reference levels (e.g., reference category for sex, iron status, and at baseline time).
- β_1: The effect of Age_i (time-invariant) on the log-odds of having hemoglobin $\leq 10\,\text{g/dL}$, accounting for age-related differences in anemia risk between individuals.
- β_2: The effect of Sex_i (binary: 0 = male, 1 = female; time-invariant) on the log-odds of having hemoglobin $\leq 10\,\text{g/dL}$, representing how anemia risk differs by sex.
- β_3: The effect of Iron_{ij} (binary: 0 = deficient, 1 = adequate; time-varying) on the log-odds of having hemoglobin $\leq 10\,\text{g/dL}$, accounting for changes in iron status over time and their influence on anemia risk.
- β_4: The effect of LagDose_{ij} (previous EPO dose; time-varying) on the log-odds of having hemoglobin $\leq 10\,\text{g/dL}$, accounting for how past dosing influences anemia risk.

- β_5: The effect of LagHB_{ij} (previous hemoglobin levels; time-varying) on the log-odds of having hemoglobin $\leq 10\,\text{g/dL}$, representing how prior hemoglobin levels influence current anemia risk.
- β_6: The effect of Month_{ij} (time) on the log-odds of having hemoglobin $\leq 10\,\text{g/dL}$, representing how the risk of anemia changes over time for each subject.
- β_7: The interaction between LagHB_{ij} and Month_{ij}, indicating how the effect of prior hemoglobin levels on anemia risk varies over time.
- β_8: The interaction between Age_i and Month_{ij}, indicating how the effect of age on anemia risk changes over time.
- β_9: The interaction between Sex_i and Month_{ij}, indicating how the effect of sex on anemia risk changes over time.
- β_{10}: The interaction between LagDose_{ij} and Iron_{ij}, indicating how the effect of EPO dose depends on iron status.
- β_{11}: The interaction between Age_i and Sex_i, capturing combined effects of age and sex.
- b_{i0}: The random intercept, representing subject-specific deviations from the population-level intercept, accounting for individual differences in baseline anemia risk.
- $b_{i1} \cdot \text{Month}_{ij}$: The random slope for Month_{ij}, representing subject-specific trends in anemia risk over time, allowing individual trajectories to vary from the population-level trend.

Table 9: Parameter Estimates: Direct Likelihood vs. Multiple Imputation

Parameter	**Direct Likelihood**			**Multiple Imputation**		
	Estimate	SE	P-value	Estimate	SE	P-value
Intercept	1.7958	1.1706	0.1251	16.3597	2.9912	0.0002
Age	-0.0133	0.0084	0.1148	0.0023	0.0059	0.6505
Sex (Male)	-0.5684	0.5167	0.2784	-0.1011	0.2446	0.6798
Iron (Adequate)	-0.1670	0.1747	0.3570	-0.1006	0.1891	0.6037
LagDose	-0.0028	0.0008	0.0068	-0.0007	0.0007	0.3702
LagHB	-0.1928	0.0833	0.0391	-1.6928	0.2709	$< .0001$
Month	1.0595	00.4166	0.0110	-1.0821	0.9203	0.2678
LagHB*Month	-0.1325	0.0352	0.0002	-0.1151	0.0844	0.2034
Age*Month	-0.1325	0.0110	0.3696	-0.0012	0.0015	0.4288
Sex*Month	0.1364	0.0418	0.0954	0.0042	0.0427	0.9222
LagDose*Iron	0.0024	0.0012	0.0365	0.0004	0.0009	0.6573
Age*Sex	0.001381	0.0007	0.8512	0.0015	0.0009	0.6540

A comparison between the Direct Likelihood (DL) results and the results after multiple imputation (MI) is shown in Table 9. The observed differences in the significance of interactions between Direct Likelihood (DL) and Multiple Imputation (MI) can likely

be attributed to how missing data is handled in these methods. In DL, only complete cases are analyzed, which reduces the dataset size and may amplify relationships that are not robust, particularly for interactions involving variables with substantial missingness, such as dose, hemoglobin, and sex. This can lead to overestimated effects or spurious significance in DL for interactions like LagHBMonth, LagDoseIron, or Sex*Month (borderline). In contrast, MI accounts for the uncertainty introduced by missing data by incorporating imputation variability across datasets. Consequently, MI provides more conservative estimates, and the lack of significance in MI for certain interactions suggests that these effects are not strongly supported when missingness is properly addressed.

6 Sensitivity Analysis

To assess the robustness of our conclusions based on the MAR assumptions, sensitivity analysis was considered to measure the extent to which the results depend on the missing data mechanism and the assumptions made during imputation and modeling. Specifically, we wanted to explore how the inclusion of imputed data impacts parameter estimates and their significance levels on the continuous outcome.
By comparing results under varying assumptions about missing data, testing the stability of results across different imputation methods, and evaluating potential deviations from MAR (MNAR scenarios), this helps quantify the uncertainty introduced by missingness and ensures the conclusions drawn are not overly reliant on the unverified MAR assumptions. Furthermore, alternative model specifications and different numbers of imputations (m = 5,10,20) could also be evaluated to assess the robustness of results under varying imputation strategies. One drawback of sensitivity analysis is its computational expense, especially when dealing with large datasets or complex models. Due to these constraints, it was not feasible to perform a full sensitivity analysis in this study.

7 Conclusions

Overall, our dataset contains a significant amount of missing data, which substantially impacts the results of our analysis. Missingness in critical covariates and outcomes introduces considerable uncertainty, emphasizing the need for thoughtful handling of missing data. While models under the MAR assumption provide reasonable estimates, sensitivity analyses are essential to assess their robustness and explore alternatives to address potential issues with missingness.

For future analyses, we recommend improving data collection strategies to minimize missingness, particularly for key covariates, or increasing the sample size to enhance the stability and reliability of estimates. Advanced sensitivity analyses under MNAR assumptions should be incorporated to evaluate the potential effects of non-random missingness. Additionally, exploring alternative imputation techniques could better capture the data's underlying structure. These measures will help mitigate challenges posed by missing data and improve the validity and reliability of future findings.

8 Appendix

```
/*GLMM*/
```

```
data hemo_data_MI;
    set hw3.longmilmm3_lag_hbbin;
run;


proc sort data=hemo_data_MI;
    by _Imputation_ ID time;
run;

data hemo_data_MI;
    set hemo_data_MI;
    month_class = time;
run;

proc glimmix data=hemo_data_MI method=quad(qpoints=5);
    class ID _Imputation_ month_class sex iron;
    model HBBIN (event='1') = age sex iron lagdose laghb time
                                laghb*time age*time sex*time
                                lagdose*IRON age*sex
        / solution dist=binomial link=logit COVB;
    random intercept time / subject=ID;
    by _Imputation_;
    ods output ParameterEstimates=glimmix_output2 CovB=glimmix_covb2;
run;

proc print data=glimmix_output2 (obs=10);
run;

proc print data=glimmix_covb2 (obs=20);
run;


/*POOL RESULTS*/
proc mianalyze parms=glimmix_output2 covb(EFFECTVAR=ROWCOL)=glimmix_covb2 wcov
bcov tcov;
 class sex iron;
    modeleffects Intercept age sex iron lagdose laghb time
                                laghb*time age*time sex*time
                                lagdose*IRON age*sex;
run;

/*DIRECT LIKELIHOOD*/
data hemo_data;
    set hw3.hemodialysis;
run;

data hemo_data;
```

```
    set hemo_data;
    month_class = month;
run;

/* Create HBBIN for binary response */
data hemo_data;
    set hemo_data;
    if hb = . then HBBIN = .; /* if hb is missing HBBIN should too*/
    else HBBIN = (hb <= 10);
run;



data hemo_data;
    set hemo_data;
    by ID;

    retain lagHb lagDose;
    lagHb = .;
    lagDose = .;

    if first.ID then do;
        lagHb = .;
        lagDose = .;
    end;
    else do;
        lagHb = lag(Hb);
        lagDose = lag(DOSE);
    end;
run;

proc glimmix data=hemo_data method=quad(qpoints=5);
    class ID month_class sex iron;
    model HBBIN (event='1') =  age sex iron lagdose laghb month
                               laghb*month age*month sex*month
                               lagdose*IRON age*sex
        / solution dist=binomial link=logit COVB;
    random intercept month / subject=ID;
    ods output ParameterEstimates=glimmix_output CovB=glimmix_covb;
run;
```